

Hypothesis

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Hypothesis

Beyond Sprint Speed: A Dual-Barrier Model of Agility, Motor Specificity, and Neuromuscular Aging Across the Lifespan

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Abstract

Masters sprinters can retain remarkable straight-line speed into older age, creating the appearance of youthful movement despite continued biological aging. This hypothesis paper distinguishes outstanding task performance from restoration of the integrated neuromuscular system. The Dual-Barrier Model combines motor specificity with biological neuromuscular aging. First, improved straight-line speed does not automatically transfer to increasingly demanding change-of-direction tasks. Second, competitive Masters athletes can retain substantial muscle and sporting expertise while exhibiting age-associated differences in body composition, muscle quality, motor-unit organization, and power production. Human studies of rapid force development provide a further measurement window into function that maximal strength and visual observation may not capture. The model introduces whole-body reactive force-time capacity: the ability to organize, generate, absorb, redirect, and stabilize force across body segments when the required response is incompletely known in advance. It predicts that Masters sprinters will show larger age-associated differences relative to appropriately characterized young athletes as movement demands increase in multidirectional control and reactive uncertainty. Successful task completion need not preclude differences in force-time behavior, coordination, or stabilization. Existing evidence supports the component distinctions, but does not establish the proposed age-by-demand interaction in Masters sprinters. Body composition is one contributor, not a sufficient explanation, and residual impairment cannot automatically be assigned to neural degeneration. No evidence reviewed here demonstrates that training restores the integrated neuromuscular system of an older adult to its young-adult biological state. A preregistered, within-participant comparison of sprinting, planned change of direction, and matched reactive tasks is proposed to test or falsify the central prediction.

Keywords: masters athletes; motor specificity; neuromuscular aging; reactive agility; rate of force development; sarcosthenia; body composition; motor units

Introduction

Exceptional athletic performance in later life raises a question about what preserved performance establishes biologically. An older athlete can remain highly skilled in a demanding, extensively practiced task without necessarily retaining the integrated neuromuscular state of a young adult. Masters sprinters provide the focal example: remarkable straight-line speed can coexist with continuing age-associated changes in the system that supports movement. This paper examines the limits of inferring biological youth from such performance.

The first foundation of the Dual-Barrier Model is motor specificity. Young et al. (2001) provided experimental evidence that improvements following straight-sprint training did not broadly transfer as predetermined change-of-direction demands became more complex. Their study concerned training specificity in young men; it tested neither aging nor reactive agility. Its relevance is that proficiency in straight-line sprinting cannot be assumed to represent general multidirectional movement capacity.

The second foundation is biological neuromuscular aging. Zange et al. (2026) showed that highly trained Masters athletes could preserve substantial muscle while still exhibiting major age-associated differences in power relative to muscle volume. Complementary findings in Masters track-and-field athletes link age with body composition and jump power (Alvero-Cruz et al., 2021). These studies establish biological and functional differences despite high activity; they do not directly test reactive agility or identify a single cause of its possible age-related limitations.

The purpose of the present model is to connect these foundations in a testable hypothesis: exceptional visible performance is not itself evidence of youthful underlying neuromuscular biology, and increasingly demanding multidirectional and reactive conditions may reveal age-associated differences that specialized performance makes less apparent. This proposed measurement advantage remains to be tested. The following sections distinguish the supporting evidence from that prediction and specify how it could be evaluated without equating movement success, improvement, or expertise with biological restoration.

1. The Paradox: Exceptional Performance in an Aging System

A highly trained older sprinter may run with a speed and fluency that challenge everyday expectations of aging. The performance is real. The inference that the athlete therefore retains a generally youthful movement system is a different proposition. Reviews of Masters athletes describe substantial advantages over sedentary peers alongside persistent age-associated differences, while longitudinal evidence in Masters sprinters demonstrates functional deterioration despite preservation of several measured muscle characteristics (Hendrickse et al., 2025; Pickering et al., 2021; Tanaka et al., 2020).

The same distinction applies to an accomplished dancer, tennis player, boxer, or athlete completing an intricate movement sequence. These examples illustrate an inference problem, not evidence that such individuals are incapable of complex or reactive movement. Successfully executing a task establishes that the performer met its demands on that occasion. It does not identify every biological process that produced the result, nor establish that those processes match a young-adult reference state.

The central proposition is that highly practiced performance can make continuing biological aging less apparent to an observer. Familiar tasks permit expertise, anticipation, and established coordination strategies to contribute strongly to the visible result. When movement instead demands rapid braking, redirection, stabilization, and selection among uncertain alternatives, the integrated system may be challenged in ways that a straight sprint does not assess. The proposed disproportionate age difference under those conditions is the hypothesis to be tested, not an already demonstrated property of every older athlete.

The model has two barriers. **Motor specificity** limits the inference from excellence in one movement to capacity in another. **Biological neuromuscular aging** limits the inference from excellence or improvement to restored biological youth. The first barrier concerns the relationship between task demands and practiced capacities. The second concerns the state of the biological system supporting any task. Addressing task unfamiliarity does not, by itself, establish reversal of the second barrier.

The phrase *"Sprinting can mask aging; reactive agility can expose it"* is a conceptual shorthand. "Mask" refers to an observer's inference from impressive performance, not to the absence of measurable aging in sprinting itself. "Expose" refers to a predicted measurement advantage under specified demands, not to inevitable visible failure. Sprinting is already sensitive to age and requires sophisticated neural and mechanical control (Hendrickse et al., 2025; Pickering et al., 2021). Figure 1 distinguishes the established component evidence from the proposed interaction.

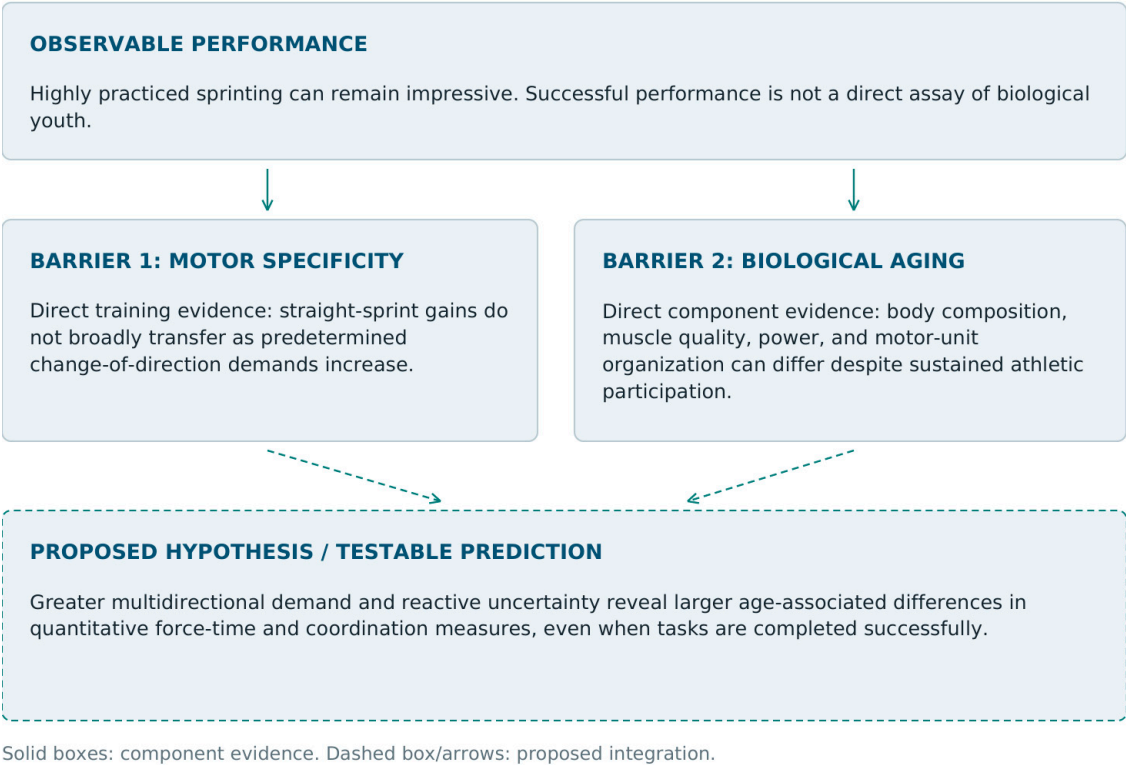


Figure 1. The Dual-Barrier Model: Performance is not a biological assay. Established component evidence supports task specificity and persistent biological differences in trained older people. The dashed connection denotes the hypothesis: increasing mechanical demand and uncertainty will reveal larger age-associated differences. The figure does not depict a proven temporal cascade. Sources: (Alvero-Cruz et al., 2021; Hendrickse et al., 2025; Young et al., 2001; Zange et al., 2026).

2. Scope, Evidence Selection, and Inferential Boundaries

This is a critical narrative synthesis and hypothesis paper, not a systematic review, meta-analysis, or report of a new human experiment. The supplied original articles and reviews were examined for their populations, methods, outcomes, numerical claims, and relevance to the model. Duplicate copies of the same paper were treated as one source. Targeted supplementary searches addressed specific gaps in human motor-unit evidence, muscle composition, calcium handling, braking mechanics, and reactive movement assessment. Selection was purposive; no exhaustive search coverage or formal certainty grading is claimed.

Four levels of statement are used throughout. **Direct evidence** describes what a cited experiment or observational study measured. **Supported inference** connects findings without claiming that the connection itself was experimentally demonstrated. **Author's hypothesis** identifies the proposed explanation. **Testable prediction** specifies an outcome that could support or weaken that explanation. A cross-sectional association is not a within-person aging trajectory, and a statistical residual is not a measured biological mechanism.

The focal comparison is between young trained athletes and older Masters sprinters, not between an exceptional older athlete and an untrained younger person. The latter comparison can demonstrate exceptional performance but cannot establish biological equivalence. Studies of non-athletic older adults, isolated muscle contractions, or upper-limb learning are included only for the component questions they actually address. Their findings are not relabeled as direct tests of whole-body reactive agility in Masters sprinters.

Citation years follow the identified journal versions. In particular, the supplied Zange paper appeared online in 2025 but carries the 2026 journal issue; it is cited here as 2026. The supplied early

version of the Ruggiero and Gruber review predates its 2026 issue assignment. These bibliographic distinctions do not denote different experiments. All numerical displays below identify the original source, the quantity plotted, and whether uncertainty is available.

3. Improvement, Preservation, Compensation, and Restoration

The model requires precise endpoints. *Improvement* means a change from an individual's baseline in a specified outcome. *Preservation* means maintenance over a defined period or better retention relative to a comparator. *Compensation* means achieving function through an alternative organization or strategy. *Restoration* would require evidence that a previously altered variable, or a prespecified integrated set of variables, returns to the relevant young-adult state. These definitions prevent a faster completion time from being treated as proof that lost motor units, contractile properties, or tissue composition have all been restored.

Young-adult biology is not a single numerical value. A credible restoration claim requires a defined reference population, comparable measurement conditions, and equivalence margins established before analysis. Matching a young group's mean on one outcome is not evidence of equivalence across the entire neuromuscular system. Conversely, a difference on one measure does not prove that every component is impaired. The present model concerns integrated function and its task dependence; it does not require uniform deterioration across all tissues.

The bounded claim is: **No evidence reviewed here demonstrates that training restores the integrated neuromuscular system of an older adult to its young-adult biological state.** This statement does not establish that every possible intervention is incapable of reversing every age-sensitive variable. Indeed, some variables can improve substantially. The critical distinction is the inferential distance between such an improvement and whole-system restoration.

The supplied high-speed resistance-training follow-up makes this distinction concrete. In 36 older completers, a 16-week intervention was followed by assessments over 12 months. Peak knee-extension and flexion RTD remained above the pre-intervention baseline. Subsequent activity groups were assigned from self-reported activity rather than randomized, and the follow-up analysis excluded the original control group. No young-adult comparison established restoration. The findings demonstrate retained gains in measured torque development, not rejuvenation of the integrated system (Duarte Martins et al., 2025).

That paper's dynamic isokinetic RTD, sampled at 100 Hz and filtered at 6 Hz, also differs from a high-resolution isometric early-RFD protocol. Very short time windows contain few samples and are sensitive to processing choices. Its results should therefore not be equated with direct measurement of restored motor-unit recruitment, young-adult early RFD, or reactive whole-body control (Duarte Martins et al., 2025; Maffiuletti et al., 2016).

4. Barrier 1: Motor Specificity

4.1. Young's Experiment and the Limits of Sprint Transfer

The foundational experiment by Young et al. (2001) enrolled 36 men, mean age 24.0 years, with prior sprinting or change-of-direction experience. Participants were randomized to straight-sprint training, change-of-direction training, or a control condition. The intervention lasted six weeks, with two training sessions per week. Twenty-seven men completed the protocol: 11 in the speed group, nine in the change-of-direction group, and seven controls. Training distances and overall volume were designed to be comparable across the two intervention groups.

Seven timed tests each covered 30 m. One was straight sprinting; the other six increased the number and/or angle of predetermined direction changes. The speed group improved significantly in the straight sprint and the simplest change-of-direction condition, which involved two 20-degree deviations. It did not show significant improvement in the more demanding conditions. The change-of-direction group improved in all six direction-change tests but not significantly in the straight sprint. Figure 2 displays changes calculated from the published group means (Young et al., 2001).

These results support a substantive boundary: straight-line sprint ability is not a general measure of multidirectional movement capacity. Training that improved linear speed did not produce comparable improvements as braking and directional requirements became more dissimilar to the trained task. This conclusion is more specific than saying merely that different training methods produce different adaptations.

The inference must nevertheless remain bounded. Within-group nonsignificance does not prove exactly zero transfer. The study had small final groups and attrition, and not every test showed a significant group-by-time interaction. It also does not establish that strength, power, or acceleration adaptations can never assist a different task. The evidence supports limited, task-dependent transfer rather than complete physiological independence (Young et al., 2001).

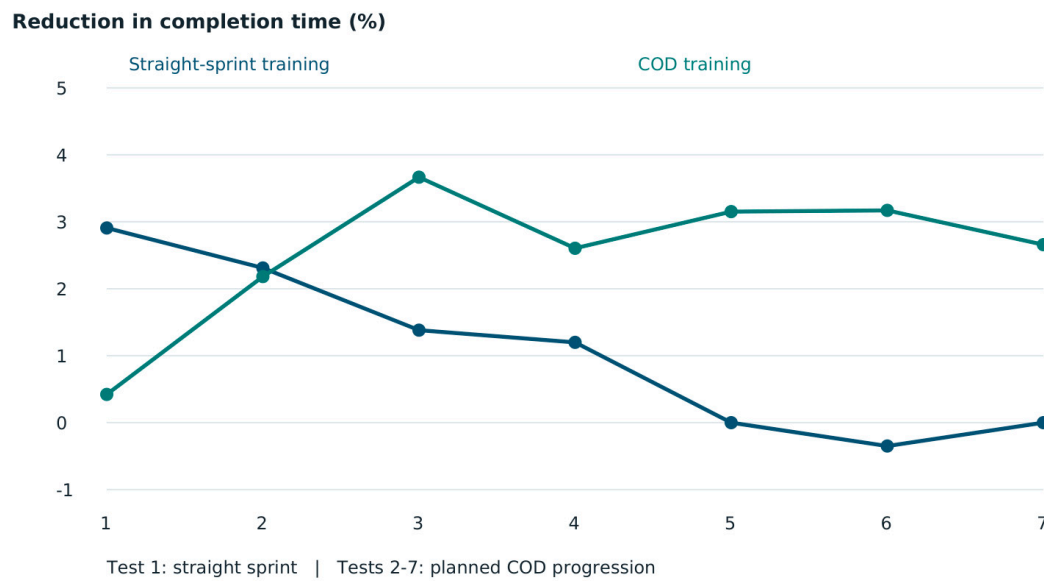


Figure 2. Straight-sprint gains did not broadly transfer to complex planned COD. Percentage reductions in time calculated as $100 \times (\text{before} - \text{after})/\text{before}$ from Table 2 of Young et al. (2001). Positive values indicate faster performance. Straight-sprint group $n = 11$; COD group $n = 9$. All courses covered 30 m. The chart shows ratios of published group means, not individual percentage changes. Error bars are omitted because paired change-score uncertainty cannot be recovered from the table. Statistical findings are described in Section 4; these are descriptive changes, not an aging or reactive-agility experiment.

4.2. Planned Change of Direction Is Not Reactive Agility

Young's courses were known in advance. They tested predetermined change of direction (COD), not stimulus-driven route selection or aging. A reactive-agility task adds a response to information that determines how movement should proceed. An open task may also allow relevant options to change during execution. These distinctions matter because a mechanical change in direction and a late requirement to select that direction impose overlapping but non-identical demands (Sheppard & Young, 2006; Young et al., 2001).

Accordingly, the model does not use Young as proof of the age-by-uncertainty interaction. Young establishes Barrier 1 in a young male sample. I propose that biological aging imposes an additional constraint when the same person moves beyond a highly practiced sprint into mechanically and informationally different tasks. The contribution of each barrier must be measured rather than inferred from a disappointing drill performance.

5. Barrier 2: Biological Neuromuscular Aging

5.1. *A Distributed System, Not a Proven Chronological Cascade*

The movement system includes cortical preparation and control, descending communication, spinal integration, motor units, neuromuscular transmission, excitation-contraction coupling, muscle-tendon mechanics, and sensory feedback. Age-associated differences have been described at multiple levels. The supplied review by Borzuola et al. (2020) synthesizes central and peripheral findings, including cortical and white-matter changes, spinal circuitry, motor-unit remodeling, and activation. It also identifies uncertainty about whether particular changes represent dysfunction, compensation, or both.

This evidence supports a distributed biological barrier. It does not establish a single chronological chain in which one cortical event necessarily initiates all subsequent muscle deterioration. Imaging-derived cortical thickness, white-matter diffusion measures, and neurophysiological excitability are different measurements; none should be translated directly into a count of lost motor units. Nor should a percentage change in neuronal cell-body volume be presented as the same percentage loss of cortical neurons or cortex. The model does not depend on the disputed simplification of a single motor-cortex percentage.

Some central findings are task dependent. A systematic review and meta-analysis of upper-extremity practice included 11 studies with 132 older and 128 young adults. Practice-related corticospinal excitability changes depended on task, while changes in excitability and inhibition did not consistently explain the magnitude of skill acquisition. This constrains any claim that one neural marker defines all movement capacity (Berghuis et al., 2017). A separate systematic review of 80 studies examined EEG oscillations during motor learning across adulthood; its heterogeneous tasks and recording paradigms likewise do not supply a single biological-youth score (Jahanian-Najafabadi & Davoodi, 2025).

These studies are used to delimit the neural explanation, not to recast the paper as a question about learning capacity. Retained learning is compatible with an aging system. The relevant issue is whether quantitative measures of the system supporting performance differ from a specified young reference under the demands of interest.

5.2. *Motor-Unit Loss, Remodeling, and Compensation*

A motor unit comprises a motor neuron and the muscle fibers it innervates. Surviving axons can reinnervate denervated fibers, preserving some contractile tissue while changing motor-unit organization. Human electrophysiological evidence therefore permits preservation of muscle function to coexist with fewer, larger motor units; successful compensation is not synonymous with restoration of the original organization (Borzuola et al., 2020; Piasecki et al., 2019).

Piasecki et al. (2019) compared vastus lateralis properties in 85 young and older men divided among controls, endurance athletes, and power athletes. Older groups had lower motor-unit number indices and larger motor-unit potentials, with greater potential enlargement in older athletes. A separate comparison of 11 Masters tennis players found larger potentials in the favored arm without a significant difference in motor-unit number indices. These observations support local remodeling and possible enhanced reinnervation with long-term use. The indices are estimates derived from electrophysiology, not direct anatomical counts. The young power group included weightlifters whereas older power athletes were sprinters, limiting a strictly discipline-matched interpretation.

Evidence is not uniform. Power et al. (2016) reported 28% more functioning tibialis anterior motor units and better transmission stability in world-class octogenarian athletes than in age-matched controls. That cross-sectional comparison supports superior preservation in an exceptional group; it cannot isolate exercise from selection or genetics, and it does not demonstrate regeneration of lost units or restoration relative to young athletes. The present model allows such preservation and does not require every trained muscle to show the same deficit.

Changes in discharge behavior also matter independently of unit number. During rapid ankle dorsiflexion, older adults exhibited lower peak torque development and lower initial motor-unit discharge frequencies than young adults. Both neural activation and contractile slowing contributed to the interpretation; the result was not a demonstration that motor-unit loss alone caused the reduction (Klass et al., 2008).

5.3. Neuromuscular Junctions and Human Specificity

Animal evidence cannot be imported wholesale into human aging. Jones et al. (2017) found notable stability of human neuromuscular-junction morphology across the sampled adult lifespan, contrasting with commonly described mouse patterns. This challenges an obligatory generalized sequence of human junction fragmentation and failure. Morphological stability does not establish identical transmission under every condition, but the model must distinguish structural observations from functional instability and motor-unit remodeling.

Barrier 2 therefore does not designate one indispensable lesion shared by all older athletes. It proposes that the integrated biological state remains age sensitive despite athletic excellence. Which components contribute most to reactive movement, and whether particular deficits are compensated in a given athlete, require direct measurement.

6. Contractile Tissue and Excitation-Contraction Coupling

The biological barrier extends beyond the nervous system. Excitation-contraction coupling links membrane excitation to intracellular calcium release and force generation; calcium reuptake contributes to relaxation. Muscle fiber composition, force per unit fiber area, calcium sensitivity, tendon properties, and activation can affect the force-time response. The early phase is not exclusively neural, and an unexplained power deficit must not be relabeled a neural percentage (Maffiuletti et al., 2016; Ruggiero & Gruber, 2026).

In mechanically skinned fibers from active older and young adults, Lamboley et al. (2015) found lower specific force and calcium sensitivity in older type II fibers, together with reduced releasable sarcoplasmic-reticulum calcium in both fiber types. These are direct human cellular observations, but isolated-fiber results do not quantify their contribution to reactive agility in a Masters sprinter. They establish plausible muscular contributors that cannot be inferred from external muscle size alone.

Calcium handling also illustrates why adaptation and restoration need separate endpoints. Hunter et al. (1999) examined young and older women and a resistance-training subgroup. Older trainees increased sarcoplasmic-reticulum calcium uptake and ATPase activity, yet electrically evoked relaxation speed did not change. The finding supports modification of a component without corresponding normalization of the measured mechanical response. It also argues against treating calcium uptake as the sole rate-limiting explanation for relaxation.

The disuse synthesis by Ruggiero and Gruber (2026) identifies reductions in explosive force that can exceed maximal-force losses, with potential contributions from activation, motor-unit discharge, twitch properties, excitation-contraction coupling, and tendon stiffness. Unloading is an experimental perturbation, not synonymous with aging. It helps identify candidate mechanisms and shows why inactivity must be measured, but cannot establish that an active older athlete's deficit is simply a failure to train enough.

Nor is slowing obligatory in every sample. In the longitudinal Masters-sprinter study discussed below, reduced performance was interpreted primarily in terms of diminished force-generating capacity rather than additional slowing. Biological aging can therefore remain evident without every candidate mechanism changing in parallel (Hendrickse et al., 2025).

7. Body Composition Belongs Within Barrier 2

7.1. Alvero-Cruz: Active Aging Does Not Mean Unchanged Composition

Alvero-Cruz et al. (2021) investigated competitors at the 2018 World Masters Athletics Championships in Málaga. The parent field study recruited 256 athletes; the methods describe 240 participants in the present substudy, aged 35-91 years, with a mean age of 58 ± 12 years. Body composition was estimated using an InBody S10 segmental multifrequency bioelectrical-impedance analyzer, and vertical-jump performance was measured using a Leonardo force platform. This was a cross-sectional field study, not longitudinal follow-up of individual athletes.

The authors reported an age-related increase in body-fat percentage of **2.0 percentage points per decade, with a standard error of 0.4 percentage points**. They described close agreement with previous longitudinal observations in Masters distance runners. They also judged the age-related changes in percent fat and fat-free mass in these highly active competitors to be comparable to those previously reported in the general population. That comparison concerns age-related patterns; it does not mean identical absolute composition, or no benefit from training. The study did not recruit a concurrent general-population control group (Alvero-Cruz et al., 2021).

Its Table 1 reports a male-cohort median body-fat percentage of **18.4%** and a female-cohort median of 21.1%. The male value is not a mean for sprinters, not a value for athletes over 70, and not a population norm. The paper contains inconsistent sample totals across the abstract, methods, and descriptive tables; its Table 1 headings give 162 men and 91 women. Those reporting inconsistencies should not be concealed by assigning a single denominator to every analysis. Only five participants were over 80 and 31 over 70, limiting inference about extreme old age (Alvero-Cruz et al., 2021).

Crucially, age remained associated with peak jump power after adjustment for athletic specialization, percent fat-free mass, and phase angle. The authors interpreted this residual age association as compatible with sarcopenia, or deteriorated muscle quality. Their model explained substantial variation, but it did not identify the biological cause of the remaining age effect. Bioimpedance-derived fat-free mass is not a direct measure of contractile protein or a map of tissue lipid distribution, and phase angle is not a motor-neuron assay (Alvero-Cruz et al., 2021).

This supports the paper's central distinction: high-level competitive participation and preserved sporting capability do not establish unchanged body composition or youthful power. It does not demonstrate that extra fat alone explains a reactive-movement deficit. Figure 3 places the body-composition result alongside the muscle-volume/power evidence without treating either as a direct agility experiment.

7.2. The Same Fat Percentage Does Not Mean the Same Tissues

Whole-body fat percentage is a ratio: fat mass divided by total mass. It does not specify where the fat is located, how much contractile tissue is present in a particular muscle, or the mechanical properties of that tissue. Consequently, the same percentage at 25 and at 70 or older cannot, by itself, establish equivalent muscle composition or equivalent function. Equal percentages also need not imply equal absolute fat mass when total body mass differs.

Human imaging evidence supports examining tissue composition separately. In 2,627 adults aged 70-79 in the Health ABC baseline analysis, computed-tomography muscle attenuation was associated with strength independently of muscle cross-sectional area. Attenuation is an indirect marker related to lipid content, not a direct assay of neural function (Goodpaster et al., 2001). Zange's MRI and spectroscopy findings in athletes further distinguish muscle-level fat fraction from whole-body adiposity (Zange et al., 2026).

The relevant compartments include subcutaneous and visceral adipose tissue, intermuscular fat, fat within a muscle boundary, and lipid within muscle cells. These are not interchangeable quantities. Nor is all intracellular lipid evidence of pathological tissue replacement. In Zange's study, age- and exercise-related differences were attributed predominantly to extramyocellular rather than

intramyocellular lipid in the spectroscopically examined soleus. Whole-body bioimpedance cannot resolve those distinctions (Zange et al., 2026).

An athletic silhouette therefore cannot serve as evidence that a Masters athlete has retained young-athlete composition. Equally, a whole-body fat measurement should not be used to diagnose myosteatorsis or infer motor-unit loss. The model requires composition and function to be measured separately.

7.3. Mechanical Implications Without a Fat-Only Explanation

For a given movement trajectory, greater mass requires greater net force to achieve the same acceleration and greater impulse to achieve the same change in velocity. At the same entry speed, additional mass also increases the kinetic energy that must be managed during braking. Adipose tissue contributes to the mass being moved without actively producing skeletal-muscle force. This is a mechanical inference; it does not establish the magnitude of a body-fat effect in a particular athlete. Braking performance additionally depends on force application, absorption, technique, and the time and distance available (Harper et al., 2022).

$$F_{net} = m a; \quad J_{net} = \int F_{net} dt = m \Delta v; \quad E_k = \frac{1}{2} m v^2.$$

These expressions assume constant body mass during the movement. Force and velocity changes are vectors. They describe mechanical requirements, not a fitted aging model or a prescription for a body-fat target. Body-fat percentage alone is insufficient to calculate the relevant forces; approach speed, body mass, muscle capacity, segment geometry, and strategy must also be known.

The supplied obesity study provides adjacent evidence. Ferhi et al. (2024) compared 20 older adults with obesity and 22 controls. The obesity group had poorer rapid plantar-flexor force measures and slower walking despite no significant difference in absolute maximal force. This was an observational comparison of older care-center recruits, not Masters athletes or a fat-loss intervention. It supports assessing rapid force relative to the load moved, but does not establish that fat accumulation caused the difference.

Interpretation also requires caution: the paper inconsistently labels its early-RFD interval, states an age nonsignificance despite different reported group means, and prints the walking-speed equation in reversed form. Its directional findings can inform the discussion, but its early-window effect is not used as a precise quantitative foundation for the present model. No inference about an athletic body-fat threshold is drawn (Ferhi et al., 2024).

8. Zange: Preserved Muscle Is Not Preserved Youthful Power

Zange et al. (2026) provides a central test of the assumption that athletic muscle preservation implies youthful function. The cross-sectional sample comprised 43 men: 10 young athletes aged 20-35, 10 Masters athletes aged 60-75, 12 young controls, and 11 older controls. Athletes competed in sprinting or jumping events. Six-point Dixon MRI quantified volume and fat fraction in 17 hip and leg muscles; soleus spectroscopy distinguished lipid compartments, and a countermovement jump assessed peak power.

The Masters athletes generally had more muscle than older controls and less muscle fat than older controls, while their muscle fat fractions exceeded those of young athletes. The summed gluteal, quadriceps, and triceps-surae volumes were 5,758 ± 1,139 mL in young athletes and 5,285 ± 895 mL in Masters athletes. These results support substantial muscle preservation alongside persistent age-associated compositional differences (Zange et al., 2026).

The decisive functional result was an age-dependent power-volume relationship. Regression slopes were **0.89 W/mL in the combined young groups and 0.45 W/mL in the combined older groups**. When estimated lean volume was used, the respective slopes were 0.88 and 0.46 W/mL. Thus, adjustment for the imaged fat fraction did not eliminate the age difference. The authors summarized

peak power per muscle volume as approximately halved in the older groups. The plotted values in Figure 3 are regression slopes across people, not a claim that each Masters athlete generates exactly half a young athlete's power (Zange et al., 2026).

The authors decomposed the athlete-group power difference into volume loss, increased muscle fat, and a remaining volume-independent component described as sarcosthenia. In that calculation, volume and fat together accounted for approximately 36% and the residual for approximately 64% of the difference. This is a model-based partition using group values and a regression slope. It is **not a directly measured 64% neural contribution**, nor proof of a single intrinsic molecular mechanism. The discussion's printed kW/mL unit is inconsistent with Table 6 and the calculation; W/mL is the applicable unit (Zange et al., 2026).

Sarcosthenia is used here in the sense employed by these papers: reduced muscle functional quality beyond quantity alone. A lower whole-body power-to-muscle-volume relationship can include activation, contractile characteristics, tendon behavior, coordination, and measurement effects. Calling the residual sarcosthenia describes an unresolved functional component; it does not identify its cause.

The study was small, male-only, cross-sectional, and subject to athletic self-selection. It did not measure reactive agility, randomize lifelong exercise, or establish the best possible response to every training strategy. Nevertheless, it directly challenges the inference that more preserved muscle and sustained competition necessarily imply youthful power production. Together with Alvero-Cruz, it supports a biological barrier that includes body composition without reducing the barrier to body fat alone.

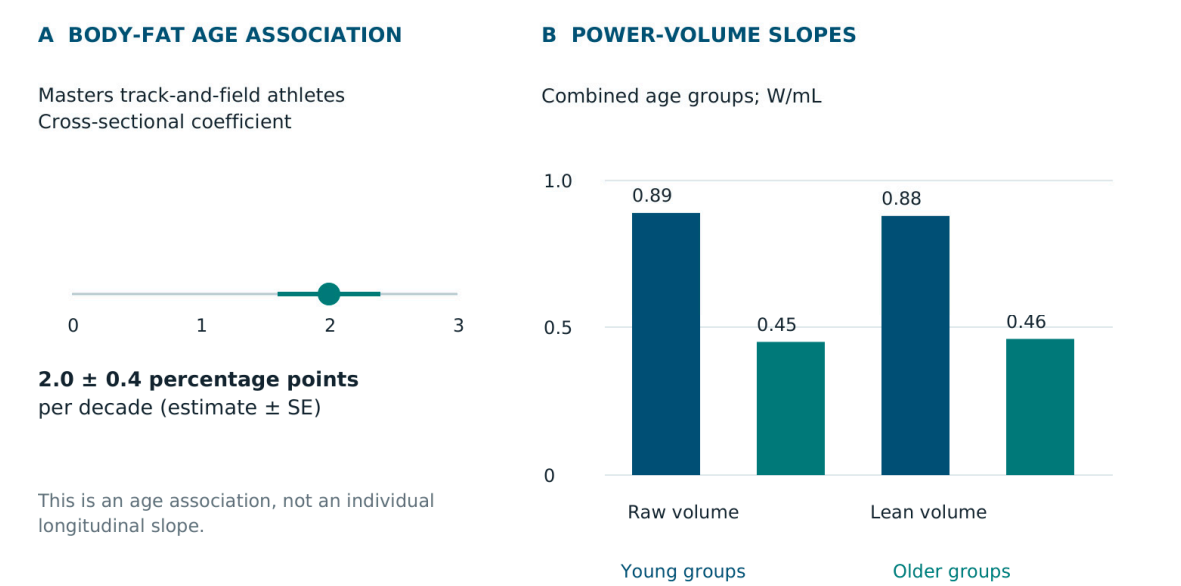


Figure 3. Body composition and muscle quality capture different parts of aging. Left: cross-sectional body-fat age coefficient reported by Alvero-Cruz et al. (2021); whisker is ± 1 standard error, not a longitudinal trajectory. Right: power-volume regression slopes from Table 6 of Zange et al. (2026), with young athlete/control groups combined ($n = 22$) and older groups combined ($n = 21$). Raw-volume slopes are 0.89 and 0.45 W/mL; estimated lean-volume slopes are 0.88 and 0.46 W/mL. No slope confidence intervals are plotted. These are distinct studies and outcomes, not a joint causal model.

9. Longitudinal Masters-Sprinter Evidence

Cross-sectional comparisons cannot exclude cohort differences. The 10-year study by Hendrickse et al. (2025) adds within-person evidence. Twenty-four male Masters sprinters, aged 48-85 at baseline, underwent muscle biopsy, ultrasound, knee-extension strength testing, jumping

assessment, and timed sprinting. Over follow-up, mean 60-m sprint time increased by 14.2% (SD 12.4), maximal voluntary knee-extension force fell by 21.0% (SD 20.4), and the reported jump-power measure fell by approximately 9.5% (SD 5.7).

At the same time, vastus lateralis thickness, average fiber cross-sectional area, and fiber-type proportions showed no statistically significant decline. This is not proof of exact structural equivalence; the sample and biopsy impose limits. It is nevertheless important evidence that retained measured morphology and continued sprint participation can coexist with declining functional output. The authors interpreted the power reduction primarily as reduced force-generating capacity, rather than additional slowing inferred from their force-velocity analysis (Hendrickse et al., 2025).

Training was not held constant. Reported weekly training fell from 7.10 to 4.67 hours, and the study could not precisely separate training changes from aging. Attrition, selection of continuing athletes, and sampling one muscle also limit inference. The absence of a significant statistical relationship between training volume and functional change does not prove that training had no influence (Hendrickse et al., 2025).

This longitudinal result supports the performance-versus-biology distinction while also constraining the slogan. Sprint performance itself deteriorated; sprinting does not literally make aging invisible to measurement. What may be concealed is the extent and distribution of biological change when an observer judges only the impressiveness of a specialized performance. The additional claim that reactive force control reveals a disproportionately larger age difference still requires the direct experiment.

10. RFD as a Measurement Window

10.1. Force Reached and Time Taken Are Different Outcomes

Maximal strength describes the greatest force attained under specified conditions. Rate of force development describes how rapidly force changes. A person can retain substantial maximal force while developing it more slowly, which matters when the available time is brief. RFD is generally expressed as a force-time slope; RTD is the corresponding joint-torque quantity. Neither is interchangeable with power, which depends on force and velocity (Maffiuletti et al., 2016).

$$RFD_{0-50} = [F(t_0 + 0.050\text{ s}) - F(t_0)] / 0.050\text{ s}.$$

Here t_0 is force onset defined by a stated detection rule. The interval is not measured from presentation of a visual stimulus. Stimulus-to-EMG latency, electromechanical delay, force-onset-to-target-force time, and total movement time are distinct intervals. Combining them without reporting their definitions would prevent a mechanistic interpretation.

Maffiuletti et al. (2016) emphasize the importance of rapid voluntary activation, especially motor-unit discharge behavior, during approximately the first 50-75 ms of an explosive contraction. Contractile properties and mechanical transmission also contribute. “Early RFD” is therefore not “pure cortical drive,” a direct count of surviving motor units, or a complete measure of reaction time. Figure 4 locates it within the broader response.

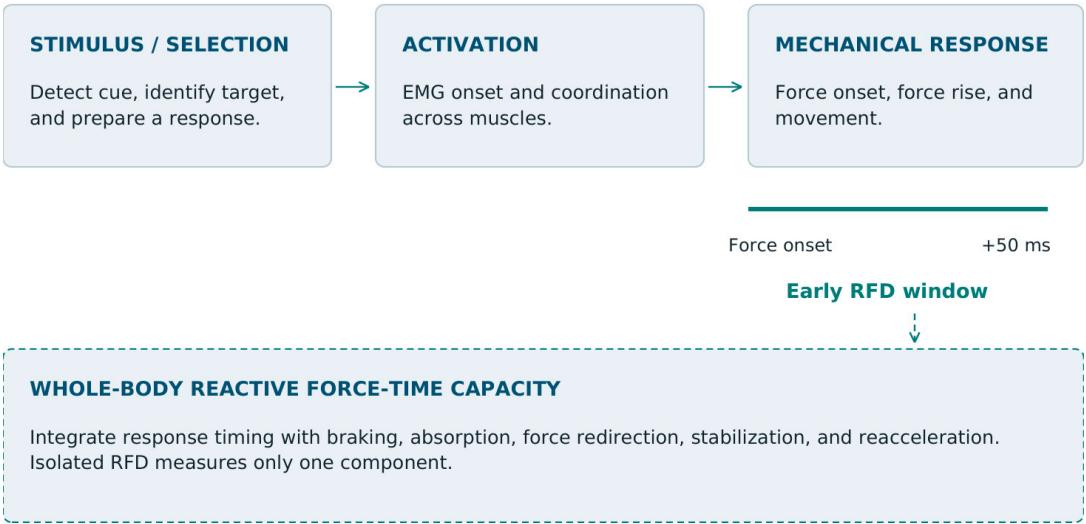
10.2. Human Aging Evidence and Limits

The ankle-dorsiflexor study by Klass et al. (2008) reported approximately 48% lower maximal RTD during fast contractions in older than young adults, accompanied by lower initial motor-unit discharge frequencies. Voluntary RTD was more affected than the electrically evoked twitch response, supporting an activation contribution alongside altered contractile behavior. This is mechanistic evidence from a constrained task, not a universal percentage applicable to whole-body movements or Masters athletes.

The supplied systematic review included 32 studies and 1,087 participants across aging and several neurological conditions. Its sample-weighted summaries showed larger RFD than maximal-strength deficits, but the associations of RFD and maximal strength with physical function were

similarly modest. The heterogeneity of populations, normalization procedures, contraction modes, and RFD windows prevents a universal claim that RFD is always the superior functional predictor. It remains a valuable additional measurement, not a replacement for all strength and movement assessments (Lomborg et al., 2022).

These observations support the measurement argument: successful movement and retained strength can coexist with altered force-time characteristics. They do not establish that every older specialist has reduced early RFD in every trained muscle. Nor can an isolated voluntary isometric RFD measurement be assumed to represent eccentric braking after foot contact. The proposed experiment measures both rather than treating them as interchangeable.



Conceptual event map. Horizontal distances are not measured durations.

Figure 4. Early RFD is one window within a whole-body response. Conceptual timing diagram, not measured duration data. The 0-50 ms window begins at detected force onset, not stimulus onset. Detection, selection, activation, and mechanical events overlap; the diagram separates measurements rather than establishing a rigid serial process. Physiological basis: (Harper et al., 2022; Maffiuletti et al., 2016). Whole-body reactive force-time capacity is a proposed construct.

11. Whole-Body Reactive Force-Time Capacity

Whole-body reactive force-time capacity is introduced here as a proposed multidimensional construct: the ability to organize and express appropriately directed force across body segments when the available advance preparation is incomplete or unpredictably altered. It includes rapid generation and absorption of force, but also the timing and coordination needed to direct that force toward a successful response. It is not an established single clinical score or an already validated biological-age index.

A reactive maneuver can involve stimulus detection, perception, response selection, motor command, activation, force generation, braking, force absorption, redirection, stabilization, and reacceleration. These processes overlap and interact with continuing feedback. The list is a functional decomposition, not a claim that movement proceeds through strictly serial, non-overlapping stages. RFD addresses part of the force-production interval; it does not quantify the entire sequence.

Horizontal deceleration has force-application and force-attenuation requirements distinct from simply running faster. A sharper turn or shorter stopping distance can change the demands even without changing uncertainty. Conversely, making a cue unpredictable can alter preparation without changing the final route. These manipulations should therefore be separated. A slower, visually

complicated task is not necessarily more mechanically demanding than a maximal sprint (Harper et al., 2022; Sheppard & Young, 2006).

Table 1 translates the construct into candidate measurements. Some have established uses in biomechanics or neuromuscular assessment; their inclusion here does not mean that each has already shown the predicted age interaction. Variables such as jerk, force-direction reversal time, and stabilization require task-specific definitions, reliability testing, and interpretation. Less variability is not automatically better: constrained or excessively rigid movement can also be less variable.

Table 1. Operationalizing whole-body reactive force-time capacity.

Domain	Candidate measurement	Interpretive boundary
Detection and selection	Stimulus-to-EMG or movement latency; wrong-direction responses; anticipatory starts	Specify the event and threshold. Latency is not RFD.
Rapid activation and force	EMG rise and intermuscular timing; RFD over prespecified force-onset windows	EMG is not a direct motor-unit count. Isometric RFD is a component test.
Braking and absorption	Braking force-time slope; eccentric impulse; center-of-mass velocity reduction	Control approach velocity and stopping distance. Ground-reaction loading is not identical to voluntary RFD.
Redirection	Horizontal and mediolateral force; force-vector reversal time; trajectory error	Define axes, turn geometry, and reversal criterion before analysis.
Coordination and stabilization	Inter-limb timing; trunk motion; time to stabilization	Strategy differences need not indicate failure; thresholds require validation.
Movement quality	Dimensionless smoothness/jerk measures; trial-to-trial variability	Control duration and path. Lower variability or jerk is not universally superior.
Task outcome	Completion time; errors; missed targets; reacceleration	Analyze speed and accuracy together. Successful completion does not imply identical kinetics.

Measurement framework proposed here. Component physiology and braking rationale: (Harper et al., 2022; Maffiuletti et al., 2016). The table does not claim that every listed metric has already shown the predicted age-associated decline.

12. Reactive Uncertainty as a Stress Test

The model predicts that reducing advance knowledge limits the contribution of a fully prepared response and increases the need to organize the movement online. Uncertainty can concern cue timing, target identity, direction, sequence, or changes introduced after movement begins. An operational definition must state which of these is manipulated. “Reactive chaos” is used here as descriptive language for unpredictable task demands, not as an established diagnostic category.

An experienced performer may still extract useful cues and anticipate probabilities. No laboratory task removes all prior knowledge: participants know the apparatus, the response options, and the testing context. Researchers should randomize relevant contingencies, monitor anticipatory movements, and quantify the preparation time available. Otherwise apparent age differences could reflect different strategies or familiarity rather than the intended uncertainty manipulation.

Adjacent human evidence makes this approach plausible without establishing the Masters-sprinter prediction. A 2026 dual-center study of 231 community-dwelling older adults assessed a Random Star Run with reactive direction changes. Its age- and site-adjusted discrimination of retrospective fall history exceeded that of the Timed Up and Go test. It did not compare young and Masters sprinters, did not establish prospective prediction, and used different court dimensions

across sites. It supports the possible added information from integrated reactive assessment, not proof that agility specifically exposes biological aging (Giesche et al., 2026).

Another study assessed 32 non-athletes, aged 31-84, using forward bipedal hopping, strength, and transcranial magnetic stimulation. Associations implicated both strength and a corticospinal measure, supporting assessment beyond one peripheral variable. The task was not a reactive COD test, and stimulation was assessed through a hand muscle rather than the hopping muscles. It therefore supplies a component association, not a causal explanation of whole-body reactive movement (MacKenzie et al., 2025).

The observation to be explained remains successful performance with potentially altered underlying kinetics. An older athlete can reach the correct target and appear fluent yet use a longer preparatory interval, different braking strategy, or different intermuscular timing. Those are candidate explanations and outcomes to measure, not deficits that can be diagnosed from a video alone.

13. The Central Prediction and What Would Count Against It

Principal prediction. Relative to appropriately characterized young trained athletes, Masters sprinters will show increasingly evident age-associated differences as tasks add demanding multidirectional force control and reduce advance preparation. The predicted differences concern prespecified force-time and coordination outcomes as well as completion time. They may remain detectable even when both groups successfully complete the task.

The strongest test is an interaction rather than an older-versus-younger main effect. If older participants are slower by a similar proportional amount in every condition, the experiment demonstrates an age difference but not the proposed disproportionate exposure under increasing demand. Likewise, a difference caused entirely by unequal approach velocities, unfamiliar instructions, injuries, or fear would not establish the claimed integrated mechanism.

The model does not predict that every older person performs worse than every younger person, that every agility metric becomes progressively worse, or that no specialist can compensate effectively. It predicts a group-level pattern under controlled conditions with a defined reference population. Baseline variation and overlap should be displayed rather than hidden behind group averages.

A precise absence of the age-by-demand interaction would weaken the model's central prediction. So would disappearance of the effect after balanced familiarization, or a finding that a young-adult reference distribution adequately describes the older group's prespecified integrated outcome profile. A poorly measured null result would be inconclusive. Investigators must specify reliability, equivalence bounds, and what size of interaction would be scientifically meaningful before seeing the data.

Evidence that one cellular or behavioral variable remains preserved narrows the proposed mechanism. It does not automatically test the whole-system prediction. Conversely, the model must not escape falsification by moving from one unfavorable metric to another after results are known. Table 2 specifies the different evidentiary roles of the core studies.

Table 2. Evidence supporting and constraining the model.

Source / design	Direct finding relevant to this paper	What remains unestablished
Young et al., 2001 Randomized training	36 enrolled; 27 male completers. Six weeks. Sprint gains transferred to the simplest but not broadly to more complex planned COD tests.	No aging comparison; no reactive route selection; limited power for small effects.
Alvero-Cruz et al., 2021 Cross-sectional field composition and age independently related study	Masters championship cohort; methods describe 240 adults aged 35-91. Body to jump power.	Variable sample totals; no imaging of lipid compartments; no direct reactive test.

Zange et al., 2026 Cross-sectional MRI	43 men in four groups. Older athletes preserved substantial muscle; the age difference in power-volume slopes persisted after fat correction.	Residual sarcosthenia is not a measured neural mechanism. Small, selected sample.
Hendrickse et al., 2025 10-year follow-up	24 male Masters sprinters. Force, jumping, and sprint performance declined without significant change in several sampled morphological measures.	Training volume changed; no reactive comparison; no single established cause.
Piasecki et al., 2019 Electrophysiology	Older athletes had motor-unit remodeling; larger potentials were compatible with compensatory reinnervation.	Indices are not anatomical counts. Cross-sectional athlete selection and discipline differences.
Power et al., 2016 Octogenarian comparison	World-class athletes showed better motor-unit preservation and transmission stability than age-matched controls.	Does not separate training from selection/genetics or establish restoration to young biology.
Lamboleylet al., 2015 Human single fibers	Older type II fibers had reduced specific force/calcium sensitivity; releasable calcium was reduced in both fiber types.	No attribution of whole-body agility loss to these cellular differences.
Klass et al., 2008 Rapid dorsiflexion	Lower voluntary RTD accompanied lower initial motor-unit discharge in older adults. RFD deficits exceeded maximal-strength deficits in heterogeneous populations;	A constrained muscle task is not a whole-body reactive assay.
Lomborg et al., 2022 Systematic review	functional associations were modest for both.	No universal superiority of RFD; no Masters-sprinter uncertainty interaction.
Ferhi et al., 2024 Older obesity comparison	Poorer rapid plantar-flexor force and walking speed despite similar absolute maximal force.	Observational; not athletes; reporting inconsistencies limit numerical inference.
Duarte Martins et al., 2025 Training follow-up	RTD gains remained above older participants' baseline over 12 months.	No young comparator or whole-system restoration endpoint; activity groups not randomized.
Giesche et al., 2026 Reactive assessment	Reactive direction-change assessment discriminated retrospective fall history in older adults beyond a conventional mobility test.	No young/Masters comparison; no prospective fall prediction or full force-time profile.

Direct findings support component distinctions. None of these studies alone demonstrates the proposed age × mechanical-demand/reactive-uncertainty interaction in Masters sprinters. Review context and additional constraints are discussed in the text.

14. The Direct Experiment

14.1. Participants and Comparison Strategy

Recruit young trained sprinters and Masters sprinters, with age modeled continuously and descriptive age strata specified in advance. Young adults aged approximately 20-35 and Masters groups spanning later adulthood would provide a useful initial design; these are proposed recruitment ranges, not data. Recruit both sexes, record sex-specific competition and training characteristics, and power the analysis for the interaction rather than only a main group difference.

Document event specialization, years and continuity of training, current training load, multidirectional sport exposure, injury, pain, vision, relevant medication, and habitual physical activity. Sprint specialists should not be presumed inexperienced in other movements. Add a multidirectional-athlete comparison group if feasible to distinguish task expertise from age-associated biology, but retain the young-versus-Masters sprinter contrast as the primary question.

Matching needs care. Matching on sex, relevant training history, and health can improve interpretability. Strictly matching on sprint speed may select unusually fast older athletes and relatively slower young athletes, creating selection bias and answering a different question. Report an age-group comparison across the recruited performance range and a separately labeled performance-overlap analysis. Do not treat age-graded rankings as biological equivalence.

14.2. Task Progression with Separable Demands

Test the same participants in straight sprinting, simple planned COD, more demanding predetermined multidirectional courses, matched reactive COD, and an open reactive condition with changes introduced during execution. Hold surface, footwear requirements, travel distance where feasible, and measurement procedures constant. Provide equivalent familiarization, counterbalance order, and monitor fatigue.

The progression alone is insufficient because complexity, speed, and uncertainty can change together. Embed matched task pairs: the same route cued early versus late, and the same directional geometry under different stopping demands. Record actual approach velocity; do not assume that equal verbal instructions produce equal mechanical exposure. Include both a standardized absolute-speed condition and a performance-relative condition if feasible, with their distinct interpretations prespecified.

Randomize cue timing and direction independently within defined safe bounds. Log cue timestamps and all target changes. Record false starts, anticipatory weight shifts, wrong-direction responses, missed targets, and incomplete trials. Analyzing only successful trials could remove the most informative failures and distort the age comparison. Figure 5 summarizes the design.

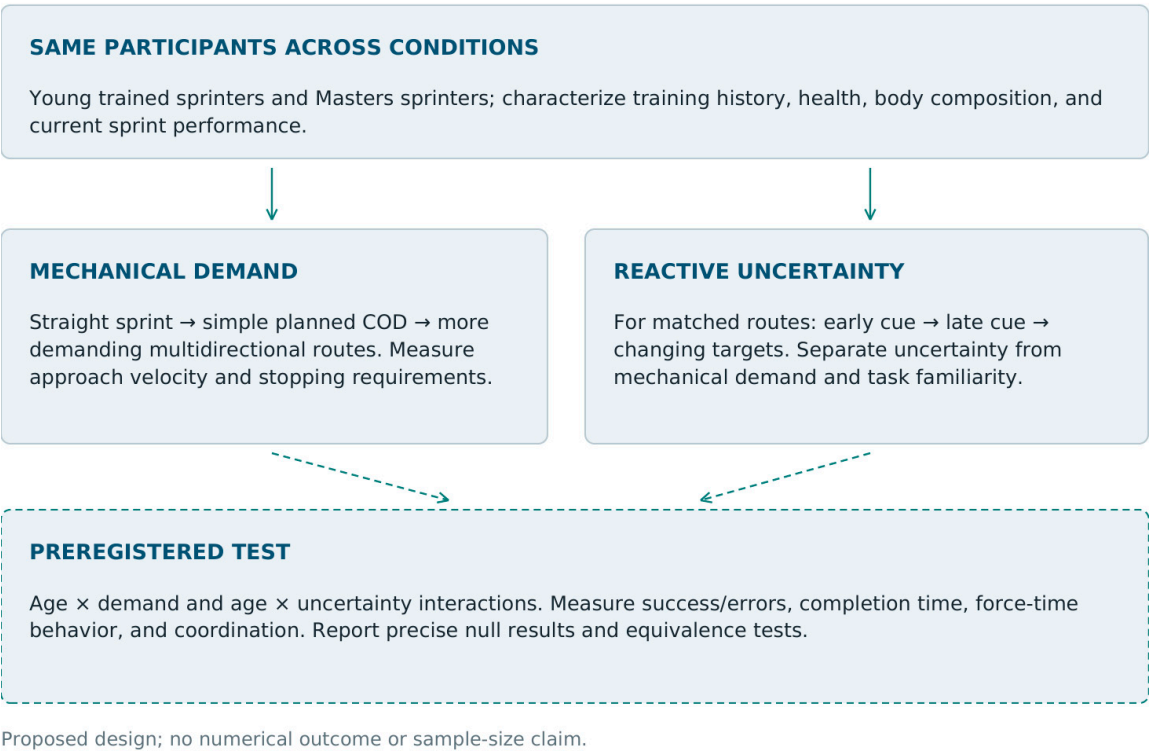


Figure 5. A direct test of the age-by-demand prediction. Proposed experiment; no participant data or effect sizes are depicted. Test the same young and Masters sprinters across tasks, with matched planned/reactive pairs separating mechanical demand from uncertainty. Quantify errors, completion time, and underlying force-time behavior. A precise absence of the predicted interaction would weaken the model. Measurement rationale: (Harper et al., 2022; Maffiuletti et al., 2016); adjacent reactive-assessment evidence: (Giesche et al., 2026).

14.3. Instrumentation and Biological Characterization

Synchronize force plates, three-dimensional motion capture or validated high-speed video, and EMG to the stimulus clock. Capture the relevant contacts through the whole maneuver rather than measuring only a single convenient footfall. Prespecify force onset, braking onset, force-vector reversal, target attainment, and stabilization. Collect repeated trials so that reliability and within-person variability can be estimated.

Include separate standardized assessments of maximal force and early isometric RFD. These quantify component capacity; the movement trials assess expression under uncertainty. Record body mass and body composition. DXA can characterize regional lean and fat mass; MRI in a mechanistic subgroup can assess specific muscle volumes and fat fractions. Avoid calling DXA lean tissue pure muscle or treating MRI fat fraction as identical to whole-body fat percentage. Bioimpedance, if used for feasibility, requires standardized hydration and explicit acknowledgment of its limitations (Alvero-Cruz et al., 2021; Zange et al., 2026).

A mechanistic subgroup could add motor-unit decomposition during standardized contractions, electrically evoked responses, and selected corticospinal measures. The purpose would be to test associations between components and the reactive outcome, not to infer that an EMG amplitude directly counts neurons. Muscle biopsy or calcium assays would require separate justification and cannot be replaced by a performance residual (Borzuola et al., 2020; Lamboley et al., 2015; Maffiuletti et al., 2016).

Safety screening and progressive familiarization are essential for maximal braking and unpredictable movement in older participants. They should limit risk without systematically giving one group a different test. Report exclusions and inability to attempt a condition, since the resulting sample may represent especially robust Masters athletes.

14.4. Analysis and Falsification

Select one primary movement outcome before data collection, such as stimulus-to-target completion time for matched reactive and planned conditions, with errors reported alongside time. Choose a small set of key secondary kinetic outcomes, for example braking impulse, time to reverse the horizontal force vector, and stabilization time. Early RFD and biological measurements provide mechanistic secondary outcomes. Do not collapse all variables into a newly weighted composite after inspecting the results.

A mixed-effects model can account for repeated trials within people and estimate interactions of age with mechanical demand and cue uncertainty. For time outcomes, logarithmic modeling can distinguish proportional from merely absolute differences. Separate scales and tasks may require separate models rather than an artificial common unit. The following is a proposed analysis structure, not an equation fitted to existing observations:

$$Y_{it} = \beta_0 + b_i + \beta_{AA_i} + \beta_{DD_t} + \beta_{UU_t} + \beta_{ADAD_t} + \beta_{AUAU_t} + \gamma X_{it} + \varepsilon_{it}$$

Y is a specified outcome for participant i in trial t; A is age, D mechanical task demand, U uncertainty, b a participant-specific intercept, and X prespecified covariates such as measured approach velocity and trial order. Appropriate nonlinear terms, random slopes, and any three-way interaction should be preregistered and supported by the sample size. The directional prediction concerns age-related costs of demand and uncertainty; it does not assign numerical coefficient values.

Estimate the total age-by-demand pattern first. Then use staged analyses adding body composition, strength, and rapid-force measures to examine how much of that pattern they statistically account for. These variables may lie on the proposed pathway, so automatic adjustment can remove part of the effect of interest. Cross-sectional attenuation after adjustment is not causal mediation. A longitudinal or randomized component would be needed for stronger causal attribution.

Use pilot estimates of reliability and within-person covariance for simulation-based sample-size planning. No definitive sample size is claimed without those inputs. Report confidence intervals,

individual distributions, missingness, sex-related heterogeneity, and sensitivity to performance matching. Table 3 states outcomes that would support, qualify, or weaken the prediction.

Table 3. Prespecified tests and potentially falsifying results.

Prediction	Supporting observation	Weakening or qualifying observation
Disproportionate age cost	Age-associated differences increase under matched mechanical or uncertainty demands.	Precise absence of the interaction, despite adequate challenge and reliability.
Difference beneath task success	Correctly completed trials show prespecified age-associated kinetic or timing differences.	Young and Masters groups meet equivalence criteria on the prespecified integrated measurement set.
Composition is contributory	Composition statistically accounts for some, but not all, of the age-by-demand pattern.	Composition alone accounts for the pattern; the proposed additional mechanism requires revision.
Beyond simple unfamiliarity	Differences persist after balanced familiarization and adjustment for relevant exposure.	The interaction disappears after familiarization or is confined to unequal experience.
Mechanistic relevance	Rapid-force or biological measures add reproducible explanatory value across participants.	No incremental value with adequate precision; proposed component pathways are weakened.
Specificity of uncertainty	Late cues add an age-related cost when route and approach velocity are controlled.	Apparent reactive differences are fully attributable to mechanical or speed differences.

Proposed tests, not completed findings. Equivalence margins, primary outcomes, and multiplicity control must be specified before data collection. A nonsignificant result alone is not evidence of equivalence.

15. Evidence That Constrains the Mechanism

A rigorous model must survive evidence of selective preservation. The supplied visuomotor study found preserved implicit internal-model recalibration despite age differences in overall adaptation. That result contradicts a blanket claim that older adults necessarily lose all predictive motor mechanisms. Its arm-reaching paradigm does not establish preservation of rapid, loaded whole-body redirection, but it narrows the neural explanation the present paper can defend (Vandevoorde & Orban de Xivry, 2019).

Similarly, upper-limb learning and EEG studies show that altered neural measures do not map one-to-one onto behavioral outcomes. Their proper role here is to prevent the inference that aging makes learning impossible or that any particular cortical signal proves global decline. They neither demonstrate integrated rejuvenation nor directly falsify the proposed Masters-sprinter interaction (Berghuis et al., 2017; Jahanian-Najafabadi & Davoodi, 2025).

The supplied lifespan synthesis emphasizes relationships between movement, cognition, and experience across development and aging. It motivates measuring lifelong motor exposure, but does not demonstrate that learning a skill in childhood permanently preserves youthful whole-body force control, or that later learning restores that state. Exposure history is a plausible modifier to measure, not a third proven biological law (Leisman et al., 2026).

Athletic preservation itself must remain visible in the argument. Masters athletes can have superior physiological function relative to sedentary peers; Power's motor-unit findings and the preserved morphology in Hendrickse's longitudinal cohort are substantive examples. They rule out portraying every age-sensitive component as equally deteriorated regardless of activity. At the same time, Zange and Alvero-Cruz show why those benefits cannot be promoted into a claim of universally

young muscle function or composition (Alvero-Cruz et al., 2021; Hendrickse et al., 2025; Power et al., 2016; Tanaka et al., 2020; Zange et al., 2026).

Finally, task expertise may preserve performance even under uncertainty. If experienced older multidirectional athletes show little additional uncertainty cost, that would qualify the generality of the hypothesis and identify a potentially important compensatory strategy. It would not be scientifically acceptable to dismiss a carefully measured result merely because the athlete is skilled. The experiment must distinguish preservation of the measured outcome from claims about unmeasured structures, in both directions.

16. Limitations and Implications

The evidence base is assembled from different muscles, tasks, age ranges, and study designs. No combination of those papers substitutes for testing the same Masters sprinters across the proposed demand progression. Many athletic studies involve selected men and small samples; women, the oldest competitors, and people unable to continue sport are underrepresented. Cross-sectional residuals can contain unmeasured health, cohort, training, or measurement effects.

The term “biological youth” is deliberately operational rather than metaphysical. This paper does not claim to measure every aspect of aging, establish an irreversible ceiling for every individual, or derive a lifespan limit. It distinguishes a specified young-adult reference state from a successful task outcome. A positive age-by-demand interaction would support the movement hypothesis; it would not, by itself, prove permanent impossibility of restoration by every intervention.

Body composition is part of the proposed explanation, but fat-only reasoning is inadequate. Changes in the mass moved, the amount and distribution of contractile tissue, muscle lipid, and force per unit tissue interact with neural control, contractile kinetics, tendon behavior, sensory information, and strategy. Even a lean older athlete can have age-associated functional differences; a higher fat percentage alone cannot identify their cause (Alvero-Cruz et al., 2021; Goodpaster et al., 2001; Zange et al., 2026).

The practical implication is measurement discipline. A sprint score, athletic appearance, successful box jump, or completed reactive drill should not be treated as a comprehensive assay of youthfulness. Conversely, a single poor unfamiliar trial should not be treated as proof of degeneration. Interpretation requires repeated, instrumented, appropriately controlled comparisons and explicit endpoints.

17. Conclusion

A Masters athlete can retain extraordinary skill and performance without retaining a young biological movement system. Young's specificity experiment establishes that straight-sprint gains do not broadly generalize as predetermined direction-change demands increase. Human Masters-athlete studies demonstrate that persistent athletic participation can coexist with age-associated differences in body composition, power, and muscle functional quality. Rapid-force and motor-unit evidence supplies plausible component mechanisms while also showing that preservation and deterioration are heterogeneous (Alvero-Cruz et al., 2021; Hendrickse et al., 2025; Maffiuletti et al., 2016; Piasecki et al., 2019; Young et al., 2001; Zange et al., 2026).

The Dual-Barrier Model proposes that practiced, predictable performance can obscure those differences at the level of visual impression, whereas quantitative testing under increasing multidirectional and reactive demands will make them more evident. An older athlete need not visibly fail for that prediction to be supported. Nor does an improvement establish restoration of the integrated system.

The decisive next step is a preregistered comparison of the same young and Masters sprinters across matched planned and reactive tasks, separating mechanical demand from uncertainty and measuring force-time behavior as well as success. A precise age-by-demand interaction would support the hypothesis; a precise absence of that interaction under reliable, adequately challenging

conditions would weaken it. That experiment, rather than athletic appearance or an isolated training response, should determine whether reactive movement exposes biological aging beyond what sprint performance reveals.

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Data and Materials Availability: No new participant-level dataset was generated. Numerical displays are derived from the cited publications, with the calculations or source-table locations stated in their captions. The proposed experimental design has not been conducted or registered.

Ethics Statement: This manuscript is a theoretical synthesis of publicly available scientific literature. No new human-participant, animal, or proprietary data were collected. Institutional ethics approval was not required for this work.

Conflicts of Interest: The author declares no conflict of interest.

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